ChurnAI: Calibrated, Explainable and Cost-Sensitive Customer Churn Prediction

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https://github.com/KrishKamra/Customer_Churn_Prediction

***Abstract*—Customer churn models are usually judged on accuracy or ROC-AUC alone, and then deployed with a default 0.5 threshold that ignores both class imbalance and the asymmetric cost of errors. We present ChurnAI, an end-to-end, fully reproducible churn-prediction system built on the IBM Telco dataset (7,043 customers, 26.54% churn). We first establish which behavioural differences are statistically real: Welch and Mann–Whitney tests on continuous attributes, plus χ^2 tests with bias-corrected Cramér's V on 16 categorical attributes. We then run a leakage-free tournament of six learners (ElasticNet logistic regression, Balanced Random Forest, LightGBM, XGBoost, CatBoost and a stacking ensemble). Every learner is wrapped in an identical feature pipeline and in isotonic calibration, and each is scored with 2×5-fold repeated stratified cross-validation. The decision threshold is chosen to minimise the expected cost $5\,\mathrm{FN} + 1\,\mathrm{FP}$ on out-of-fold predictions, never on the holdout. The regularised linear model wins (CV PR-AUC 0.655 ± 0.018). On an untouched holdout of 1,409 customers it reaches ROC-AUC 0.848, PR-AUC 0.651, recall 86.1% and a Brier score of 0.134. That is 67% less expected loss than doing nothing, and 41% less than a blanket retention offer. We explain predictions with SHAP and model-based counterfactuals, audit subgroup behaviour, and ship the model behind a calibrated, explainable dashboard with a SQL audit trail and a CI quality gate. All experiments ran on a single laptop CPU.**

***Index Terms*—churn prediction, probability calibration, cost-sensitive learning, SHAP, counterfactual explanations, reproducibility**

I. Introduction

Retaining an existing subscriber is typically far cheaper than acquiring a new one, which makes churn prediction one of the most widely deployed applications of supervised learning in subscription businesses [1], [2]. Yet the gap between a promising notebook and a decision system that a retention team can trust is wide. Three issues recur in practice:

1) **Misaligned metrics.** With roughly one churner per three customers, accuracy rewards ignoring churners, and ROC-AUC hides poor precision in the region that matters [3], [4].
2) **Uncalibrated scores and naive thresholds.** Class re-weighting and boosting distort probabilities [5]. A fixed 0.5 threshold then ignores that a missed churner costs far more than a wasted retention offer [6].
3) **Opacity and leakage.** Target-derived encodings and preprocessing fitted before the train/validation split inflate offline metrics. Black-box scores without explanations are also hard to act on.

TABLE I
Summary of Key Results

Champion (selected by CV PR-AUC)	ElasticNet Logistic Regression
Holdout ROC-AUC / PR-AUC	0.848 / 0.651
Recall at $t = 0.185$	86.1% (322/374)
Brier score	0.134
Expected loss: model vs. no action	\$30,650 vs. \$93,471
Strongest categorical driver	`Contract` ($V = 0.410$)
Churn in the first six months	52.9%

Contributions. This report documents ChurnAI, an open-source reference system that addresses each issue:

- an **inferential analysis** that separates real churn drivers from noise before any model is trained (Section IV);
- a **leakage-free tournament** of six learners sharing one pipeline, with calibration inside every fold and a fingerprinted holdout (Section V);
- a **cost-optimal decision rule** chosen on out-of-fold predictions, plus an honest analysis of threshold sensitivity (Section VI);
- **explanations and a subgroup audit**: global and local SHAP, counterfactual retention prescriptions and subgroup error rates (Sections VII and VIII);
- a **production system**: dashboard, SQL audit log, tests and CI gate, all reproducible on a laptop CPU (Section IX and the Appendix).

Table I summarises the headline results.

II. Related Work

Churn modelling. Telecom churn prediction has a long history in marketing science. Neslin et al. [1] showed that modelling choices materially change the profit of retention campaigns. Verbeke et al. [2] argued for profit-driven rather than accuracy-driven evaluation, and Burez and Van den Poel [7] studied the effect of class imbalance. We follow this line and evaluate with PR-AUC and an explicit cost matrix.

Learners. We compare an ElasticNet-regularised logistic regression [8], the Balanced Random Forest [9] from imbalanced-learn [10], three gradient-boosting libraries (XGBoost [11], LightGBM [12], CatBoost [13]) and stacked generalisation [14]. All are implemented with scikit-learn [15].

Calibration and costs. Isotonic regression [16] and Platt scaling correct the distorted probabilities that boosting and re-weighting produce [5]. Calibration is assessed with the

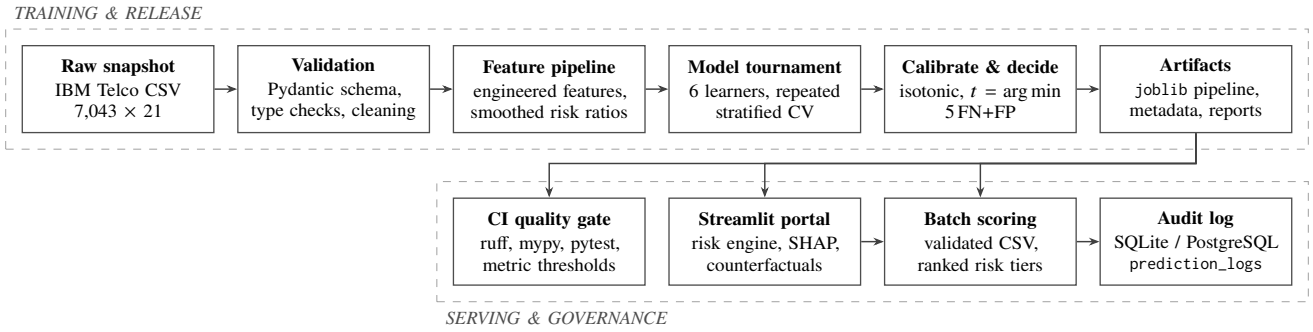


Fig. 1. The ChurnAI system. One serialized pipeline is shared by training, the CI gate, the interactive dashboard and batch scoring, so no preprocessing is ever re-implemented at inference time.

TABLE II
ATTRIBUTES OF THE IBM TELCO DATASET, GROUPED BY TYPE.

Group	Attributes	Type
Demographics	`gender`, `SeniorCitizen`, `Partner`, `Dependents`	binary
Services	`PhoneService`, `MultipleLines`, `InternetService`, `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, `StreamingMovies`	categorical
Account	`Contract`, `PaperlessBilling`, `PaymentMethod`	categorical
Billing & tenure	`tenure` (months), `MonthlyCharges`, `TotalCharges`	numeric
Label	`Churn` (Yes/No)	binary

Brier score [17]. With calibrated probabilities, the cost-optimal threshold follows directly from the cost matrix [6].

Explainability and accountability. SHAP [18] and its exact tree variant [19] attribute predictions to features. Counterfactual explanations [20] answer the question a retention manager actually asks: what would change the outcome? We document the model following model-card practice [21], and we report subgroup error rates in the spirit of equality of opportunity [22].

III. Data

We use the IBM Telco Customer Churn sample dataset [23]. It records 7,043 customers of a fictional telecom provider with 19 predictive attributes, a customer identifier and a binary label: whether the customer left within the last month (Table II). 1,869 customers churned (26.54%), an imbalance of about 1:2.8.

Cleaning. `TotalCharges` is stored as text, and 11 values are blank. All 11 belong to customers with tenure = 0 who had not yet been billed, so we set them to their true value of zero rather than imputing spend that never happened. No other values are missing.

Lifecycle risk. Churn falls steeply with tenure (Fig. 2). Customers in their first six months churn at 52.9%, versus 9.5% after four years. This *danger zone* is the single clearest operational finding in the data.

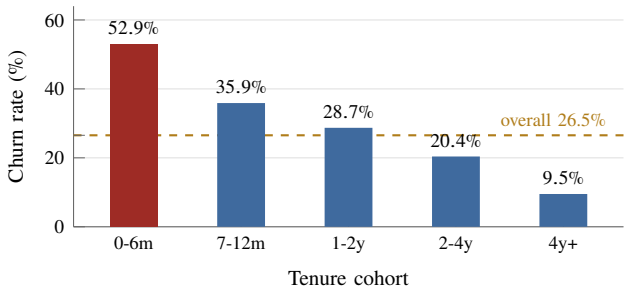


Fig. 2. Churn rate by tenure cohort. The 0–6 month danger zone (red) churns at more than twice the overall rate (dashed line).

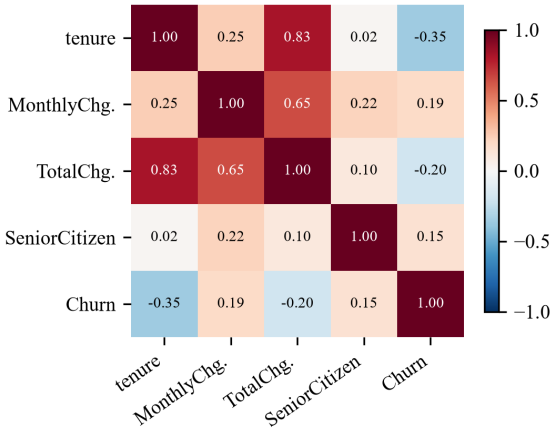


Fig. 3. Pearson correlations among the numeric attributes and the label.

Redundancy. `TotalCharges` is close to tenure times monthly charges: its correlation with tenure is $r = 0.826$ (Fig. 3). Rather than drop it, we add a *charge-velocity ratio* that keeps only the residual information, such as plan changes and discounts (Section V).

IV. Inferential Analysis

Before training models, we test which differences between churners and retained customers are statistically real, using $\alpha = 0.05$.

Continuous attributes. Normality (Shapiro–Wilk) and equal variances (Levene) are both rejected for every attribute. We therefore pair Welch's t-test [24] with the distribution-free Mann–Whitney U test [25], and report the rank-biserial correlation as an effect size (Table III). Churners have markedly shorter tenure and lower lifetime spend but *higher* monthly charges: they are new customers on expensive plans. The two tests agree on every variable.

Categorical attributes. For each of the 16 categorical attributes (the 15 string attributes plus `SeniorCitizen`) we test independence from churn with a χ^2 test. We measure association strength with Cramér's V [26], using the bias-corrected estimator of Bergsma [27]:

$$\tilde{V} = \sqrt{\frac{\tilde{\varphi}^2}{\min(\tilde{k}-1, \tilde{r}-1)}}, \qquad (1)$$

$$\tilde{\varphi}^2 = \max\Big(0, \frac{\chi^2}{n} - \frac{(k-1)(r-1)}{n-1}\Big),$$

with $\tilde{k} = k - (k-1)^2/(n-1)$, and $\tilde{r}$ defined analogously. Table IV ranks the attributes. `Contract` is the strongest driver ($V = 0.410$), followed by the absence of security and support add-ons, fiber internet and electronic-check payment. `gender` ($\chi^2 = 0.52$, $p = 0.470$) and `PhoneService` are independent of churn. This gender result anchors the subgroup audit in Section VIII.

V. Method

A. Feature pipeline

Every learner receives the same scikit-learn pipeline. Its first stage is a stateless `FeatureEngineer` that never sees the label, so it is safe to apply anywhere. It adds seven numeric features and one categorical one: $\log(1+x)$ transforms of both charge variables, the number of add-on services, fiber/month-to-month and automatic-payment flags, average monthly spend, a tenure cohort, and the charge-velocity ratio

$$v = \frac{\text{TotalCharges}}{\text{tenure} \cdot \text{MonthlyCharges} + \epsilon}, \qquad v \in [0, 5]. \qquad (2)$$

A target-aware `RiskRatioEncoder` then encodes three high-churn interactions (Internet×Contract, Contract×Payment, Cohort×Contract) as smoothed risk ratios

$$\rho(g) = \frac{1}{\pi} \cdot \frac{n_g\, \bar{y}_g + m\, \pi}{n_g + m}, \qquad m = 20, \qquad (3)$$

where π is the training-fold churn rate and $\bar{y}_g$ the churn rate of group g. Unseen groups encode to $\rho = 1$. Finally, numeric columns are median-imputed and standardised, and categorical columns are one-hot encoded with unknown levels ignored. Because the encoder and every learned transform sit *inside* the pipeline, they are re-fit on each training fold, which removes the most common source of leakage.

B. Learners

Table V lists the six contenders and their regularised defaults. Imbalance is handled by class weighting (`balanced`, `scale_pos_weight`) or balanced bootstrap sampling, never by synthetic oversampling.

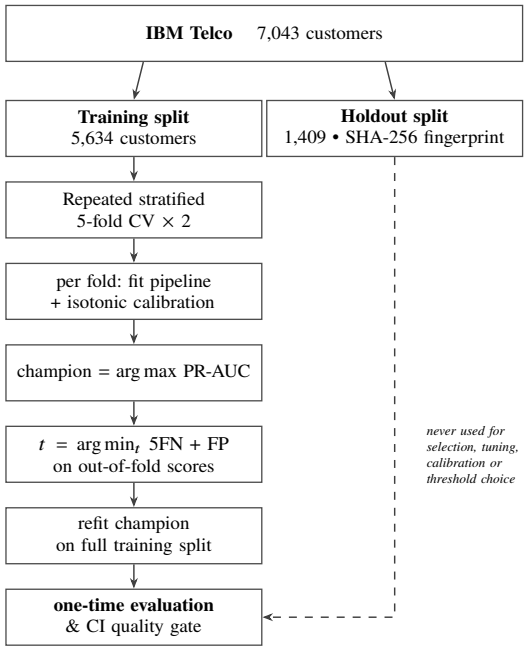


Fig. 4. Evaluation protocol. The holdout is touched exactly once.

C. Evaluation protocol

Fig. 4 summarises the protocol. A stratified 80/20 split with a fixed seed reserves 1,409 customers as a holdout. We store the SHA-256 fingerprint of the holdout customer IDs, and the CI gate refuses to evaluate if it changes. On the 5,634 training customers, each contender, wrapped in isotonic `CalibratedClassifierCV`, is scored with repeated stratified 5-fold CV (×2 repeats).

Metrics. The primary metric is PR-AUC (average precision), which is more informative than ROC-AUC under imbalance [3]. Secondary metrics are ROC-AUC, the Brier score $B = \frac{1}{n}\sum_i (p_i - y_i)^2$ [17], recall, precision, F_2, and the business cost

$$C(t) = c_{\text{FN}} \cdot \text{FN}(t) + c_{\text{FP}} \cdot \text{FP}(t), \qquad c_{\text{FN}} = 5,\ c_{\text{FP}} = 1, \quad (4)$$

where one cost unit (one wasted retention offer) is valued at \$50. This is an illustrative assumption, and it is configurable.

Decision threshold. For perfectly calibrated probabilities, the Bayes-optimal rule intervenes when $p \geq t^* = c_{\text{FP}}/(c_{\text{FP}}+c_{\text{FN}}) = 1/6$ [6]. We correct for residual miscalibration by minimising Eq. (4) over a threshold grid on the out-of-fold probabilities. We take the median across repeats, which gives $t = 0.185$ for the champion. This is close to $1/6$, which itself suggests the probabilities are well calibrated.

Tuning. After selection, a 30-iteration randomised search over the champion's hyper-parameters (here, C and the ℓ_1 ratio) is re-scored with the same CV protocol. Its result is adopted only if it improves mean CV PR-AUC. In our run it did not, so the regularised defaults were kept.

TABLE III

CONTINUOUS ATTRIBUTES, CHURNED VS. RETAINED CUSTOMERS. r_{rb} IS THE RANK-BISERIAL CORRELATION; POSITIVE VALUES MEAN CHURNERS RANK LOWER.

	Mean			p-value		
Attribute	Churned	Retained	Welch t	Welch	Mann–Whitney	r_{rb}
`tenure`	18.0	37.6	-34.8	1.2×10^{-232}	2.4×10^{-208}	+0.48
`MonthlyCharges`	74.4	61.3	18.4	8.6×10^{-73}	3.3×10^{-54}	-0.24
$\log(1 + \textit{TotalCharges})$	6.3	7.2	-19.2	9.8×10^{-78}	5.7×10^{-83}	+0.30

TABLE IV

χ^2 TEST OF INDEPENDENCE AND BIAS-CORRECTED CRAMÉR'S V FOR ALL 16 CATEGORICAL ATTRIBUTES, RANKED BY EFFECT SIZE. THE LAST COLUMN SHOWS THE ATTRIBUTE LEVEL WITH THE HIGHEST CHURN RATE.

#	**Attribute**	χ^2	dof	p-value	**Cramér's V**	Sig.	**Highest-risk level**
1	`Contract`	1,184.6	2	5.9×10^{-258}	0.410	✓	Month-to-month (43%)
2	`OnlineSecurity`	850.0	2	2.7×10^{-185}	0.347	✓	No (42%)
3	`TechSupport`	828.2	2	1.4×10^{-180}	0.343	✓	No (42%)
4	`InternetService`	732.3	2	9.6×10^{-160}	0.322	✓	Fiber optic (42%)
5	`PaymentMethod`	648.1	3	3.7×10^{-140}	0.303	✓	Electronic check (45%)
6	`OnlineBackup`	601.8	2	2.1×10^{-131}	0.292	✓	No (40%)
7	`DeviceProtection`	558.4	2	5.5×10^{-122}	0.281	✓	No (39%)
8	`StreamingMovies`	375.7	2	2.7×10^{-82}	0.230	✓	No (34%)
9	`StreamingTV`	374.2	2	5.5×10^{-82}	0.230	✓	No (34%)
10	`PaperlessBilling`	259.2	1	2.6×10^{-58}	0.191	✓	Yes (34%)
11	`Dependents`	189.9	1	3.3×10^{-43}	0.164	✓	No (31%)
12	`SeniorCitizen`	160.4	1	9.5×10^{-37}	0.150	✓	1 (42%)
13	`Partner`	159.4	1	1.5×10^{-36}	0.150	✓	No (33%)
14	`MultipleLines`	11.3	2	0.003	0.036	✓	Yes (29%)
15	`PhoneService`	1.0	1	0.316	0.001	✗	Yes (27%)
16	`gender`	0.5	1	0.470	0.000	✗	Female (27%)

TABLE V

TOURNAMENT CONTENDERS AND THEIR MAIN SETTINGS.

Model	**Key settings**
ElasticNet LR	`saga`, $C = 0.5$, ℓ_1 ratio 0.5, balanced class weights
Balanced RF	400 trees, min leaf 3, $\sqrt{p}$ features, per-tree undersampling
LightGBM	500 trees, lr 0.03, 15 leaves, row/col subsample 0.8, $\lambda = 2$
XGBoost	400 trees, lr 0.03, depth 4, `hist`, subsample 0.8, $\lambda = 2$
CatBoost	600 iterations, lr 0.04, depth 5, $\ell_2 = 5$, balanced weights
Stacking	CatBoost + XGBoost + ElasticNet LR → logistic meta-learner (5-fold)

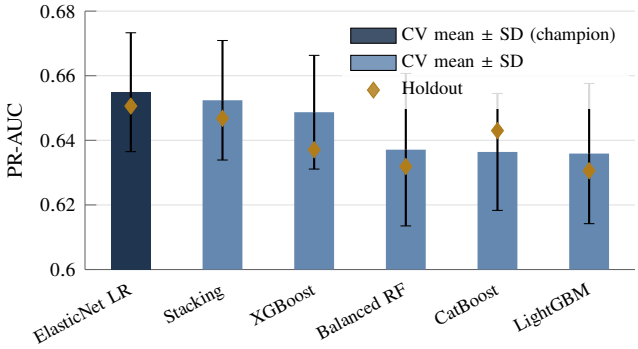


Fig. 5. Cross-validated PR-AUC (±1 SD) and holdout PR-AUC (diamonds). The differences among the top contenders lie within fold-to-fold noise.

VI. RESULTS

A. Cross-validated tournament

Table VI and Fig. 5 report the tournament. The ElasticNet logistic regression achieves the highest mean PR-AUC, but the top three contenders are separated by less than one standard deviation. On a dataset whose non-leaky ceiling is around ROC-AUC 0.85, extra model capacity buys almost nothing. The stacking ensemble ties the linear model on ROC-AUC and Brier score, while taking about 24× longer to fit.

B. Holdout performance

Table VII scores every contender once on the untouched holdout, each at its own out-of-fold threshold. The champion attains ROC-AUC 0.848, PR-AUC 0.651 and the best calibration (Brier 0.134). At $t = 0.185$ it identifies 322 of the 374 churners (recall 86.1%) at a precision of 47.7%. XGBoost records a marginally lower expected loss on this particular split. The gap (about 3%) is well within the variation between CV folds, and we keep the pre-registered selection rule rather than pick a winner on the holdout. Fig. 6 shows the champion's ROC, precision–recall and reliability curves.

TABLE VI

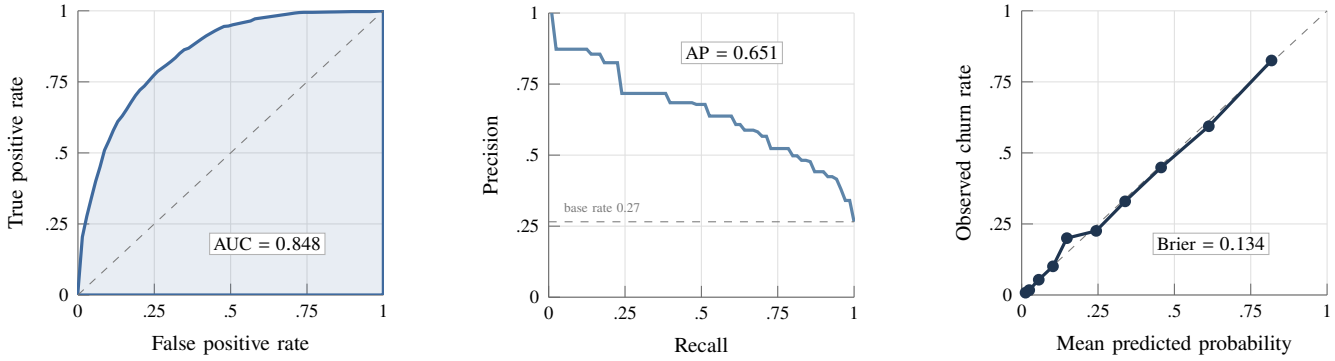
REPEATED STRATIFIED CROSS-VALIDATION ON THE TRAINING SPLIT (MEAN ± SD OVER 2×5 FOLDS; EACH FOLD RE-FITS THE PIPELINE AND CALIBRATION). COST PER CUSTOMER USES EACH MODEL'S OUT-OF-FOLD COST-OPTIMAL THRESHOLD. ⋆ CHAMPION.

Model	PR-AUC ↑	ROC-AUC ↑	Brier ↓	Cost / customer ↓	Fit (s)
ElasticNet Logistic Regression⋆	0.655 ± 0.018	0.849 ± 0.012	0.133	0.430	6.5
Stacking (CatBoost + XGBoost + LR)	0.652 ± 0.018	0.849 ± 0.012	0.133	0.427	157.3
XGBoost	0.649 ± 0.018	0.846 ± 0.012	0.135	0.430	3.8
Balanced Random Forest	0.637 ± 0.024	0.844 ± 0.013	0.136	0.438	10.0
CatBoost	0.636 ± 0.018	0.842 ± 0.012	0.137	0.444	16.5
LightGBM	0.636 ± 0.022	0.843 ± 0.013	0.137	0.442	2.1

TABLE VII

HOLDOUT EVALUATION (1,409 CUSTOMERS). EACH MODEL USES ITS OWN COST-OPTIMAL THRESHOLD t FROM OUT-OF-FOLD PREDICTIONS. LATENCY IS THE MEDIAN SINGLE-ROW `predict_proba` TIME OF THE FULL PIPELINE ON A LAPTOP CPU. ROWS ARE SORTED BY EXPECTED LOSS. ⋆ CHAMPION.

Model	t	ROC-AUC	PR-AUC	Recall	Precision	F_2	Brier	ms	Loss ($)
XGBoost	0.185	0.843	0.637	0.906	0.448	0.752	0.136	28	29,650
Balanced Random Forest	0.155	0.841	0.632	0.890	0.457	0.748	0.138	40	30,050
ElasticNet Logistic Regression⋆	0.185	0.848	0.651	0.861	0.477	0.742	0.134	21	30,650
LightGBM	0.205	0.838	0.631	0.874	0.463	0.743	0.138	26	30,700
Stacking (CatBoost + XGBoost + LR)	0.175	0.849	0.647	0.866	0.469	0.741	0.134	41	30,850
CatBoost	0.165	0.842	0.643	0.920	0.418	0.742	0.137	35	31,400



(a) ROC curve (b) Precision–recall curve (c) Reliability diagram (deciles)

Fig. 6. Holdout diagnostics for the champion. Ranking quality (a, b) is strong, and the isotonic-calibrated probabilities track observed churn rates closely (c). That is what makes the probabilities usable for cost-based decisions and for revenue-at-risk estimates.

C. Business cost and threshold sensitivity

The cost-optimal threshold transforms the model's value (Fig. 7). On the holdout, doing nothing costs $93,471 and a blanket offer to every customer costs $51,756. The champion at $t = 0.185$ costs $30,650, which is 67% and 41% lower respectively. The naive 0.5 threshold would forgo much of this gain. Table VIII shows that the loss curve is flat between $t \approx 0.10$ and 0.20. The exact threshold matters far less than moving away from 0.5. Within that flat region, the choice trades recall for retention-team workload.

TABLE VIII

THRESHOLD SENSITIVITY OF THE CHAMPION ON THE HOLDOUT. † DEPLOYED THRESHOLD (CHOSEN OUT-OF-FOLD).

t	Recall	Prec.	TP	FP	FN	Loss ($)
0.10	0.949	0.407	355	517	19	30,600
0.15	0.909	0.442	340	430	34	30,000
0.185†	0.861	0.477	322	353	52	30,650
0.25	0.816	0.498	305	308	69	32,650
0.30	0.816	0.498	305	308	69	32,650
0.40	0.671	0.588	251	176	123	39,550
0.50	0.519	0.671	194	95	180	49,750

VII. EXPLAINABILITY

Global. For linear and tree champions, SHAP values are exact (LinearSHAP or TreeSHAP [18], [19]). They are computed in log-odds space on the inner, pre-calibration pipeline. Isotonic calibration is monotone, so signs and rankings carry over to the deployed probability. We sum the SHAP values of one-hot columns back to their source attribute. Fig. 8 and Table IX show that lifecycle features dominate: the cohort×contract risk ratio, lifetime spend, average monthly spend and the tenure

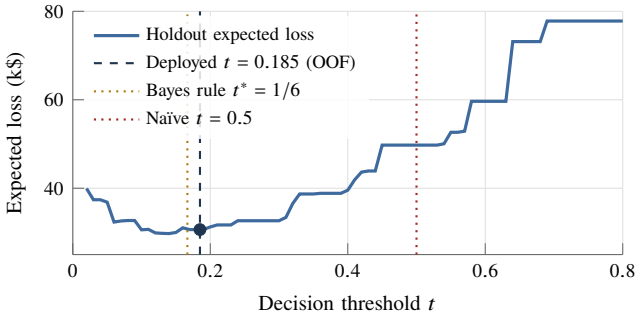


Fig. 7. Expected holdout loss as a function of the decision threshold. The deployed threshold was chosen on out-of-fold data and lies in the flat optimum, next to the Bayes rule 1/6.

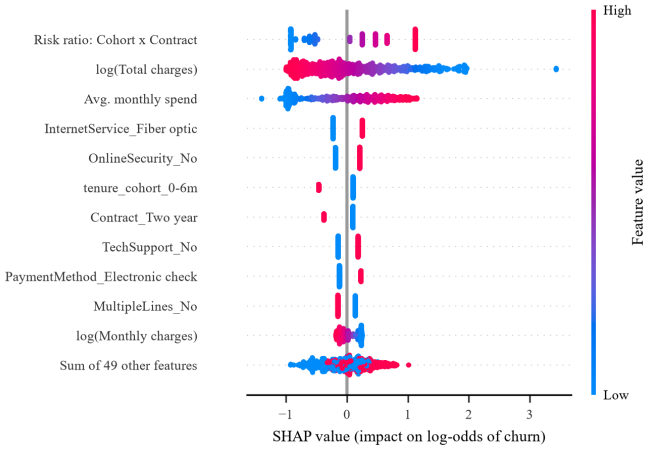


Fig. 8. Global SHAP summary on 800 holdout customers. Each dot is one customer, and the colour encodes the feature value.

cohort. Internet service, contract type and missing security add-ons follow. These are the same drivers the inferential analysis identified independently in Section IV.

Local and counterfactual. Table X explains one profile: a senior, single, fiber-optic customer two months into a month-to-month contract, paying $94.50 by electronic check, scored at 93% churn risk. The dashboard then searches a set of actionable interventions (contract migration, automatic payment, support and security bundles, a loyalty discount) and ranks the ones that lower the score. These are *model-based* what-ifs in the sense of Wachter et al. [20]: they predict the outcome for an otherwise identical customer. They are not causal estimates of an offer's uplift, which still requires controlled experiments.

VIII. Subgroup Audit

Table XI compares holdout behaviour across demographic groups at the deployed threshold. Calibration holds within every group: the mean predicted probability tracks the observed churn rate. The model behaves almost identically for women and men, consistent with the χ^2 result in Section IV. Senior citizens, however, have a much higher base rate, and under a single global threshold they receive roughly double the false-positive rate of other customers, with lower ranking quality. For a

TABLE IX
Global importance: mean |SHAP| (log-odds), with one-hot columns aggregated to their source attribute.

#	Feature	Mean |SHAP|
1	Risk ratio: Cohort x Contract	0.697
2	log(Total charges)	0.619
3	Avg. monthly spend	0.577
4	Tenure cohort	0.325
5	InternetService	0.238
6	Contract	0.231
7	OnlineSecurity	0.199
8	PaymentMethod	0.169
9	PaperlessBilling	0.169
10	TechSupport	0.163

TABLE X
Local explanation and counterfactual prescriptions for the "high-risk newcomer" profile shown in the dashboard (93% predicted churn).

Top SHAP contributions	**log-odds**
Risk ratio: Cohort x Contract = 2.03	+1.122
log(Total charges) = 5.25	+0.862
Avg. monthly spend = 94.50	+0.651
Tenure cohort = 0-6m	-0.533
InternetService = Fiber optic	+0.254
PaymentMethod = Electronic check	+0.238
OnlineSecurity = No	+0.211

Counterfactual intervention	**New p**	**Δ pts**
Migrate to a two-year contract	40%	−53
Retention bundle (annual + auto-pay + support)	40%	−53
Migrate to a one-year contract	57%	−36
Switch to automatic credit-card payment	87%	−6
Bundle Tech Support	87%	−6
Bundle Online Security	87%	−6

retention offer, a false positive means an unnecessary benefit, which is acceptable. It would not be acceptable if the score drove any adverse action. In that case, group-specific thresholds or an equal-opportunity constraint [22] would be needed. Customers with dependents see lower recall, so retention outreach may under-serve them.

IX. System

The model ships as a single `joblib` artifact with JSON metadata recording the version, threshold, metrics, holdout fingerprint, library versions and host hardware. A four-page Streamlit portal (Fig. 9) provides (i) a real-time risk engine with a calibrated gauge, local SHAP waterfall and counterfactual cards; (ii) a customer database view over SQLite or PostgreSQL, with an audit timeline of every logged score; (iii) executive analytics that recompute the inferential tests live; and (iv) schema-validated batch scoring. Inputs are validated with Pydantic. Unknown categorical levels are ignored rather than raising errors, and extra columns such as the label are dropped before scoring, which closes an inference-time leakage path. A GitHub Actions pipeline runs ruff, mypy and 60 tests, including leakage, determinism and database tests. It also re-scores the committed model on the fingerprinted holdout and fails if

TABLE XI
SUBGROUP AUDIT ON THE HOLDOUT AT $t = 0.185$. FPR = FALSE-POSITIVE RATE.

Group	n	Churn	Mean p	Recall	FPR	AUC
Female	687	28.1%	26.7%	83.4%	33.0%	0.845
Male	722	25.1%	26.8%	89.0%	35.1%	0.852
Not senior	1187	23.3%	23.9%	82.6%	30.7%	0.849
Senior citizen	222	44.1%	41.7%	95.9%	58.9%	0.784
No partner	736	33.7%	33.6%	89.1%	44.7%	0.829
Partner	673	18.7%	19.3%	80.2%	24.7%	0.852
No dependents	978	31.0%	31.9%	89.1%	42.8%	0.836
Dependents	431	16.5%	15.1%	73.2%	17.8%	0.848

ROC-AUC < 0.82, PR-AUC < 0.60, recall < 0.75 or Brier > 0.16.

X. Limitations and Ethics

Data. The dataset is a single snapshot of a fictional provider, with no time dimension. The model sees customer state, not trajectory (usage decline, support tickets, price changes), and seasonality and concept drift are not represented. Within long contracts, ranking quality is weak: at the deployed threshold, none of the two-year-contract churners in the holdout is flagged. A natural next step is longitudinal data such as the KKBox churn dataset.

Costs. The 5:1 cost ratio and the $50 unit are illustrative. Real offer costs and customer lifetime values should be substituted, and the threshold re-derived.

Causality. Counterfactual prescriptions are associational. Customers on long contracts are partly self-selected loyalists, so the predicted risk reduction overstates the effect of moving a customer onto one. Offers should be validated with A/B tests.

Use. The model is intended to prioritise *beneficial* outreach. It must not drive adverse actions such as price increases or service downgrades, nor decisions in regulated domains. Prediction logs contain customer identifiers and should be governed as personal data.

XI. Conclusion

On the IBM Telco dataset, careful protocol matters more than model complexity. A regularised logistic regression, evaluated without leakage, calibrated, and operated at a cost-derived threshold, matches or beats gradient boosting and stacking. It cuts expected loss by 67% relative to inaction, while remaining fast, exactly explainable and easy to audit. Most of the practical value comes from steps that are often skipped: confirming drivers statistically, calibrating, choosing the threshold out of sample, explaining individual decisions and auditing subgroups. ChurnAI packages these steps into a reproducible, tested and documented system.

Appendix: Reproducibility

Every number, table and chart in this report is generated from the committed model and data by `scripts/build_paper.py`. All experiments ran locally on a single laptop CPU, with no GPU or cloud compute (Table XII). The full tournament, including the 30-iteration tuning search, takes about ten minutes.

TABLE XII
COMPUTING ENVIRONMENT.

Machine	ASUS Zenbook 14 (UX3405CA), laptop
CPU	Intel Core Ultra 9 285H (16 threads)
Memory	32 GB
OS	Windows 11
Python	3.12.13, dependencies pinned with `uv.lock`
Libraries	scikit-learn 1.9.1, XGBoost 3.4.1, LightGBM 4.7.0, CatBoost 1.2.10, SHAP 0.54.1, pandas 3.1.0, NumPy 2.5.4
Seed	42 (split, CV, learners)

Commands.

```
uv sync --all-extras
uv run python -m src.models.train --tune-iter 30
uv run python -m src.models.evaluate --validate-thresholds
uv run python scripts/build_paper.py
uv run streamlit run src/ui/app.py
```

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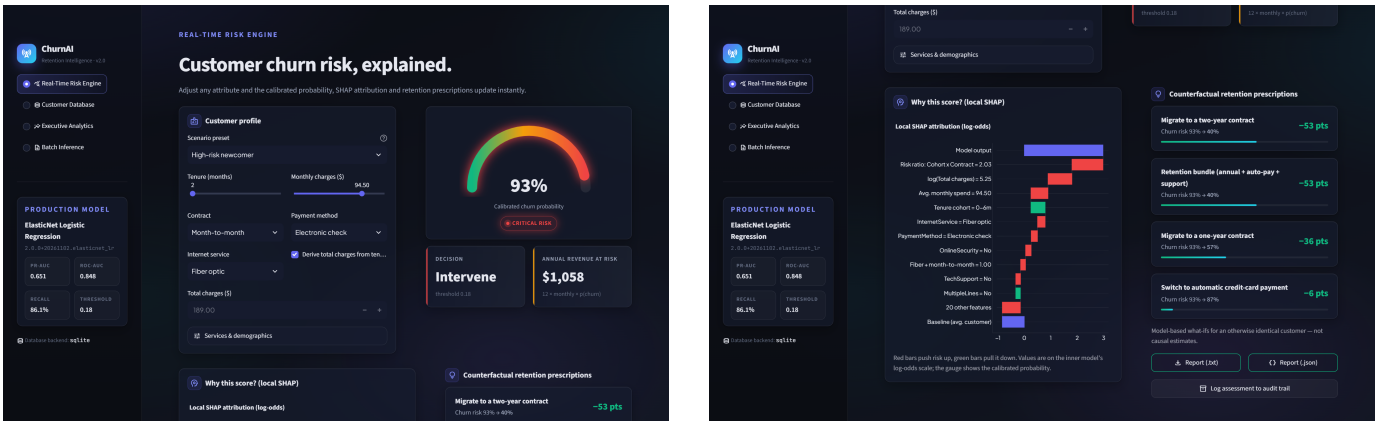
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(a) Real-time risk engine

(b) Local SHAP and counterfactual prescriptions

Fig. 9. The ChurnAI dashboard. The calibrated probability, risk tier, decision and revenue at risk update as the profile is edited. Every score can be explained, turned into a ranked list of retention actions, exported as a report, and logged to the audit trail.

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